

# MIKE (MODULE ISOGENY KEY EXCHANGE): AN $\iota\chi\theta\acute{\upsilon}\varsigma$ INTRODUCTION

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**ABSTRACT.** We give a down to earth and elementary introduction to the isogeny based cryptography protocol MIKE.

## 1. INTRODUCTION

Isogeny based cryptography often has a reputation for being too “mathematical”. In this article, we try to address this problem by illustrating in a down to earth manner the recently proposed NIKE (Non Interactive Key Exchange) MIKE (Module Isogeny Key Exchange) [Rob24]. In particular, we explain how this key exchange boils down to standard linear algebra under the hood, namely the Tensor-Hom adjunction<sup>1</sup>. Our goal in this document is to dispel this myth once and for all.

**1.1. From CSIDH to MIKE.** The best known (and first) NIKE is the Diffie-Hellman keyexchange [DH76], which works over a commutative finite group  $G$ . Unfortunately the DLP is broken by Shor’s algorithm [Sho94]. A post-quantum alternative is to instead use a commutative group action by  $G$  on a set  $X$ . An easy adaptation of Diffie-Hellman allows to build a NIKE in this setting. An instantiation is given by  $X$  the set of supersingular elliptic curves over  $\mathbb{F}_p$ , and  $G$  the class group  $\text{Cl}(R)$  of  $R = \mathbb{Z}[\sqrt{-p}]$ , which is the  $\mathbb{F}_p$ -endomorphism ring of any such curve (technically: or is of index 2 in the endomorphism ring).

This gives the CSIDH NIKE [CLMPR18]:

$$\begin{array}{ccc} E_0 & \xrightarrow{\quad} & E_{[\mathfrak{a}]} = \mathfrak{a} \cdot E_0 \\ \downarrow & & \downarrow \\ E_{[\mathfrak{b}]} = \mathfrak{b} \cdot E_0 & \xrightarrow{\quad} & E_{[\mathfrak{ab}]} \simeq \mathfrak{ab} \cdot E_0 \end{array}$$

Still, while Shor’s algorithm does not apply to a commutative group action, Kuperberg’s algorithm [Kup05] gives a quantum subexponential algorithm to recover  $g$  from  $x, g \cdot x$ .

The defunct SIDH protocol [JD11] bypassed Kuperberg’s algorithm by moving to (maximal) supersingular elliptic curves over  $\mathbb{F}_{p^2}$ . The endomorphism ring  $\text{End}_{\mathbb{F}_{p^2}}(E)$  of such an elliptic curve is a maximal order in the quaternion algebra  $B_{p,\infty}$  ramified at  $p$  and  $\infty$  (if we exclude some weird twists of the curves with  $j = 0, 1728$ ). The endomorphism ring is no longer commutative, so there is no commutative group action anymore, and Kuperberg’s attack does not apply. But then, the question was how to make a NIKE out of the  $\mathbb{F}_{p^2}$ -supersingular graph, since it has no commutative properties. The idea of SIDH was that the commutative diagram above is a pushforward diagram whenever the isogenies  $\phi_{\mathfrak{a}} : E_0 \rightarrow E_{\mathfrak{a}}$  and  $\phi_{\mathfrak{b}} : E_0 \rightarrow E_{\mathfrak{b}}$  have coprime degree. (We remark that  $\deg \phi_{\mathfrak{a}} = N(\mathfrak{a})$ , so equivalently this happens when  $\mathfrak{a}, \mathfrak{b}$  have coprime norms). Pushforward of isogenies always exist: the pushforward of  $\phi_1 : E_0 \rightarrow E_1, \phi_2 : E_0 \rightarrow E_2$  is given by  $E_{12}$ , the codomain of the isogeny  $\phi_{12} : E_0 \rightarrow E_{12}$  of kernel  $\text{Ker } \phi_{12} = \text{Ker } \phi_1 + \text{Ker } \phi_2$ . So in SIDH the shared key was the pushforward of Alice and Bob isogenies, which were furthermore taken to be of coprime degrees.

However, to enable Alice computation of this pushforward from her secret isogeny  $\phi_1$  and Bob’s codomain  $E_2$ , some extra information on Bob’s secret isogeny  $\phi_2$  was needed by the protocol, in the form of the images of some points of torsion by  $\phi_2$ . This extra information led to adaptative key attacks on SIDH, which had to be turned into the PKE SIKE; before a full attack was discovered in 2022 [CD23; MMPPW23; Rob23].

*Date:* 01/04/2026.

*Key words and phrases.* Canularichthus aprilis.

<sup>1</sup>In a suitable internal logic

The key insight of MIKE is that we can still do a key exchange in the supersingular setting over  $\mathbb{F}_{p^2}$ , but without requiring any extra information on the secret isogenies other than their codomains:

$$\begin{array}{ccc} E_0 & \longrightarrow & E_1 \\ \downarrow & & \downarrow \\ E_2 & \rightsquigarrow & A_{12} \end{array}$$

The magic trick lies in the squigly arrows to  $A_{12}$ : we use higher dimensional isogenies to build  $A_{12}$ , which is a principally polarised abelian fourfold over  $\mathbb{F}_p$ . MIKE can be seen as a natural generalisation of CSIDH to higher dimension, where we act by modules on “standard” superspecial abelian varieties over  $\mathbb{F}_p$  rather than by ideals on supersingular elliptic curves. A further trick is to use the Weil restriction of scalar functor to see supersingular elliptic curves over  $\mathbb{F}_{p^2}$  as superspecial abelian surfaces over  $\mathbb{F}_p$ , to which we can apply the module action.

**1.2. Overview.** The goal of this document is to be as pedagogical as possible, and to provide a gentle introduction to the concepts used in MIKE. For this, we follow the guidelines from the great pedagogical document [Sero3].

We will not shy from more advanced topics either, in our journey we will explore:

- Abelian varieties, i.e. elliptic curves in higher dimension, which have become a hot topic in isogeny based cryptography since the SIDH attacks.
- Polarisation on abelian varieties, which is more subtle than in the elliptic curve case (an elliptic curve has a canonical principal polarisation, so this concept stays in the background in the dimension 1 case).
- Theta functions, which allow us to work on higher dimensional principally polarised abelian varieties algorithmically.

Still, let us reassure the reader that we will not go too far into advanced topics. There will be very little quaternion algebra (and their scary non commutative nature). We will just add a bit of scheme theory for fun, in part because we need to define line bundles on abelian varieties (polarisations and projective coordinates come from line bundles), but this is mainly essentially an excuse to present the functorial view point of schemes (and we actually give several different “functorial” interpretation).

Some of the contribution of this introduction are of independent interest:

- We present a slight generalisation of MIKE, see Definition 2.2. As is often the case in mathematics, more conceptual approaches can often be simpler to understand.
- We use the fact that over an abelian variety, a line bundle (resp. a polarisation) is literally<sup>2</sup> a quadratic form (resp. a symmetric bilinear form) with values in (the delooping) of  $\mathbb{G}_m$ . This point of view considerably simplifies working with abelian varieties and their polarisations.

Because we believe that algorithms are very important to understand a mathematical concept, we also provide a high level algorithm in the appendix (Appendix A).

**1.3. Prerequisite.** Often the literature in isogeny based cryptography assume that objects like quaternion algebras or supersingular elliptic curves are well known. In this document instead, we try to assume as few knowledge as possible, and we take care to define every concept, like supersingular elliptic curves (Definition 4.5), quaternion algebras (Definition 5.1), and even what is the finite field  $\mathbb{F}_{p^2}$  (Definition 4.2).

We will only assume basic knowledge of basic category theory, like  $\infty$ -topoi and stable  $\infty$ -categories. We refer to [Lur09; Lur17; Lur18] for a short introduction to these topics. Recall that an  $\infty$ -topos is simply an accessible left exact localisation of a category of presheaf of anima. And a stable  $\infty$ -category  $\mathcal{C}$  is an  $\infty$ -category that admits finite colimits and whose suspension functor  $\Sigma$  is an equivalence. Equivalently,  $\mathcal{C}$  is a pointed  $\infty$ -category, every morphism admits a fiber and cofiber, and a triangle in  $\mathcal{C}$  is a fiber sequence iff it is a cofiber sequence. In particular, the homotopy category  $h\mathcal{C}$  is then a triangulated category, and if  $\mathcal{C}$  admits a  $t$ -structure, then its heart  $\mathcal{C}^\heartsuit$  is an abelian category.

<sup>2</sup>In a suitable internal logic

## 2. A GENERAL PURPOSE NIKE

We let  $Ani$  be the category of animae, and  $Sp = Stab(Ani)$  its stabilisation, i.e. the category of spectra. We let  $E_\infty$  be the usual little cube operads, an  $E_\infty$ -ring (or spectral ring) is then an algebra over an  $E_\infty$ -operad in the symmetric monoidal category of spectra. Recall that the sphere spectrum  $\mathbb{S}$  is the initial object in the  $\infty$ -category of  $E_\infty$ -rings;  $Sp$  is freely generated under colimits by  $\mathbb{S}$ . We denote by  $\mathbb{Z} = \pi_0(\mathbb{S})$ , the ring of integers.

A commutative group is an algebra for the commutative monad  $X \mapsto \mathbb{Z}^{(X)}$ , these form a Grothendieck abelian category  $Ab$ . This is also the heart  $Sp^\heartsuit$  of  $Sp$  for its canonical  $t$ -structure. The category  $(Ab, \otimes)$  is symmetric monoidal, and a ring is a monoid object. Equivalently, a ring is an algebra for the monad on  $Ab$  given by  $T(X) = \bigoplus_n X^{\otimes n}$ . A commutative ring  $R$  is a commutative monoid object in  $(Ab, \otimes)$ , equivalently it is an algebra for the monad  $T$  on  $Ab$  which to  $X$  associates the free polynomial ring:  $T(X) = \bigoplus_n \text{Sym}^n X \simeq \mathbb{Z}[X]$ , equivalently by the Eckmann–Hilton argument, it is a monoid in the symmetric monoidal category of rings (for the tensor product  $\otimes$ ). Finally, we let  $R\text{-mod}$  the category of  $R$ -modules over  $R$ , i.e. the category of algebras for the commutative monad  $X \in Ab \mapsto R \otimes X$ .

The category  $R\text{-mod}$  of modules over  $R$  is a strongly locally presentable category (because it is algebraic), generated under 1-sifted colimits by  $R\text{-mod}^{sfp}$ , the projective modules<sup>3</sup> of finite presentation (where  $R\text{-mod}^{sfp}$  denote the objects strongly of finite presentation in  $R\text{-mod}$ ). The category  $Ani(R\text{-mod})$  of animated  $R$ -modules is the free  $\infty$ -sifted completion of  $R\text{-mod}$  [CS24]. This is the prestable  $\infty$ -category  $D_{\geq 0}(R\text{-mod})$  (with a forgetting functor to animae), and its stabilisation is the stable  $\infty$ -category  $D(R\text{-mod})$  of derived  $R$ -modules (with a forgetting functor to spectra). If  $M$  is in  $R\text{-mod}$ , we will identify it with its animation  $Ani(M) \in Ani(R\text{-mod})$ .

The above construction makes it clear that animated modules are the category free modules really want to build, when they are liberated of the oppression of 1-categorical colimits, and allowed to soar free as birds in the  $\infty$ -categorical world.

Now let  $(X, \tau)$  be a 1-site, and  $\text{Sh}_\tau(X, R\text{-mod})$  the topos of sheafs of  $R$ -modules. By [Lur18],  $D(\text{Sh}_\tau(X, R\text{-mod}))$  is the hypercompletion of the  $\infty$ -topos of  $\text{Sh}_\tau(X, D(R\text{-mod}))$ .

**Definition 2.1.** Given a sheaf of spectra  $A \in D(\text{Sh}_\tau(X, R\text{-mod}))$  and an animated module  $M$ , then we define the derived/animated contravariant module action as  $M \cdot A = \mathcal{R}Hom(\underline{M}, A)$ , where  $\underline{M}$  is the hypersheafification of  $M$ , and  $\mathcal{R}Hom$  is the internal Hom.

**Definition 2.2** ( $\otimes^L$ -MIKE). Fix a  $R$ -sheaf of spectra  $E_0$ . The NIKE  $\otimes^L$ -MIKE works as follow: Alice picks up an animated  $R$ -module  $M_1$  and she sends  $M_1 \cdot E_0$ . Bob picks up an animated  $R$ -module  $M_2$  and he sends  $M_2 \cdot E_0$ . The shared secret key is then  $(M_1 \otimes_R^L M_2) \cdot E_0 = M_1 \cdot M_2 \cdot E_0 = M_2 \cdot M_1 \cdot E_0$ .

**Remark 2.3.** Of course, a similar NIKE works for modules over an animated ring (or even spectral ring) in an  $\infty$ -topos. We made the choice to use the standard derived Tensor-Hom adjunction to keep our protocol well grounded in basic objects.

More generally, one can build a NIKE from any action category  $C \times A \rightarrow A$  from a symmetric monoidal category  $C$ .

**Remark 2.4.** In [Rob24], a non derived module action is used. This has several drawbacks, for instance when composing the action of  $M_1, M_2$ , we need to be careful to pick them to be Tor-independent, and in particular the action of a torsion module is badly behaved. As we will see, an abelian variety is bidual to itself, so we want to act by a category with a well defined duality; again the derived dual  $RHom_R(M, \mathbb{G}_m)$  is better behaved than the standard dual  $Hom_R(M, \mathbb{G}_m)$  for a non projective module.

We will instantiate this protocol to the fppf sheaf topos over a finite field  $\mathbb{F}_{p^2}$ , and take  $E_0$  a supersingular elliptic curve over  $\mathbb{F}_{p^2}$ . As promised, since we do not assume any knowledge beyond the basic concepts, we are going to define these terms.

## 3. SCHEMES

**Definition 3.1.** A scheme is a connective 0-truncated spectral scheme [Lur18, § 1.1].

For convenience, since spectral schemes might not be in the standard curriculum, we give several alternative definitions below. From these definitions, the reader will have no trouble reading

<sup>3</sup>We can restrict to free modules of finite type, because projective modules of finite presentation are given by the Cauchy completion of free modules

EGA [GD67, Volumes I, II, III, IV], SGA [Gro71; Gro68; DG70b; AGV72; Gro77; BGI71; Gro72] and the stack project [Stacks], which from now on we will add as prerequisites.

**3.1. Schemes as Deligne-Mumford stacks.** Let  $R$  be a commutative ring. We denote by  $\mathrm{Sp}_{\mathrm{et}}(R)$  the locally ringed topos given as the classifying topos for strict Henselizations of  $R$ . Explicitly this is the small étale topos of  $\mathrm{Spec} R$ . We call these topoi affine étale topoi.

**Definition 3.2.** A ringed topos  $(X, \mathcal{O}_X)$  is said to be a DM-stack if it has a cover by objects  $U_i$  (i.e.  $\coprod U_i \rightarrow 1$  is an epimorphism), such that each  $(X_{U_i}, \mathcal{O}_X|_{U_i})$  is equivalent to an affine étale topos [Lur18, Definition 1.2.4.1].

If  $(X, \mathcal{O}_X)$  is a DM-stack, we identify it with the genuine 1-stack on the big étale topos given by the restricted Yoneda embedding  $R \mapsto \mathrm{Hom}(\mathrm{Sp}_{\mathrm{et}} R, X)$ . Then we can see a DM-stack as a stacky quotient  $[U/R]$  by schemes (or more generally algebraic spaces), where  $R \rightrightarrows U$  is an étale groupoid.

A DM-stack is said to be an algebraic space if its inertia  $I_X$  is trivial. Equivalently,  $X_{U_i \times U_j} \rightarrow X_{U_i} \times X_{U_j}$  is a monomorphism. Finally,  $X$  is a scheme if the  $U_i$  are furthermore subterminal, i.e.  $U_i \rightarrow 1$  is a monomorphism.

**3.2. Schemes as sheafs in the big étale topos.** We can also directly define an algebraic space as a sheaf in the big étale topos who has an étale cover by affine schemes (seen as étale sheafs).

However, when we adopt the sheaf point of view rather than the 1-topos point of view to define algebraic spaces, we need to explain what it means for a morphism  $\phi : F \rightarrow G$  of sheafs to be étale. This is done by remarking that an étale morphism is a formally étale morphism locally of finite presentation, and both conditions admit a functorial characterisation (unique lifting of nilpotent thickenings and commuting with inductive limits respectively). This allows us to bootstrap the definition of an étale morphism to representable morphisms of sheafs. (Using a two step bootstrap process for representability of morphisms: either we already have constructed schemes and we can go from representable by schemes to representable by algebraic spaces; or we go from representable by affine schemes to representable by algebraic spaces with affine diagonal to representable by algebraic spaces. See also [Stacks, Tag 03XV], [Stacks, Tag 04S5].) In other words, an algebraic space is a quotient  $[U/R]$  by an étale equivalence relation  $R \rightrightarrows U$ , for  $R, U$  two schemes (or algebraic spaces).

An algebraic space  $X$  is then said to be a scheme if we can find an étale cover  $U_i \rightarrow X$  where the maps are étale monomorphisms (because étale monomorphisms are open immersions).

**3.3. Schemes as sheafs in the big Zariski topos.** One can also define a scheme by working in the Zariski topos. Recall that the category of commutative rings  $\mathcal{C}Ring$  is locally finitely presentable, because it is an algebraic theory. The (big) Zariski topos  $\mathrm{Sh}(\mathcal{C}Ring_{\mathrm{fp}}^\circ, I_{\mathrm{Zar}})$  is the geometric theory of local rings. (We restrict to  $\mathcal{C}Ring_{\mathrm{fp}}^\circ$  to have a small site; technically this means that our notion of schemes is well behaved mainly for schemes locally of finite presentation, but this won't matter in the text since all our schemes will be lfp.)

In that case, we see an affine scheme  $\mathrm{Spec} R$  as a Zariski sheaf, via the Yoneda embedding as usual. Then a scheme  $X$  is a sheaf for the big Zariski topos which has a cover by affine schemes  $U_i$ , such that each  $U_i \rightarrow X$  is an open immersion. We need to define what it means for a morphism  $F \rightarrow G$  of sheaves to be an open immersion; as for algebraic spaces we can bootstrap this for representable morphisms (via a two step process to affine schemes to schemes with affine diagonals to schemes). An alternative, that I learnt from Toën's master course on algebraic stacks, is to use that an open immersion is a monomorphism. We say that a morphism of sheaves  $\phi : F \rightarrow \mathrm{Spec} R$  is an open immersion if it is a monomorphism and there exists a family of open immersions  $\mathrm{Spec} R_i \rightarrow \mathrm{Spec} R$  such that the image of  $\phi$  is equal to that of the morphism  $\coprod \mathrm{Spec} R_i \rightarrow \mathrm{Spec} R$ . Then we say that a morphism of sheaves  $\phi : F \rightarrow G$  is an open immersion if this is the case for the pullback  $F \times_G \mathrm{Spec} R \rightarrow \mathrm{Spec} R$  for any morphism  $\mathrm{Spec} R \rightarrow G$ . Note that in this case we do not require  $\phi : F \rightarrow G$  to be representable, so the pullback might not be an affine scheme.

But in fact, there is an even better approach using the subobject classifier, that I learnt from David Madore. The subobject classifier  $\Omega$  of the Zariski topos sends a commutative ring  $R$  to the set of its ideals. We have canonical subobjects  $0, 1 \rightarrow \Omega$  given by the zero ideal and the unit ideal respectively. A map  $Y \rightarrow X$  is said to be a closed, resp. open, immersion if it is given by pullback of  $0 \rightarrow \Omega$ , resp.  $1 \rightarrow \Omega$ . A presheaf  $X$  on  $\mathcal{C}Ring_{\mathrm{fp}}^\circ$  is then a scheme if it is a sheaf for the Zariski

topology, and is covered by affine opens: there exists an epimorphism  $\coprod \text{Spec } A_i \rightarrow X$  such that each  $\text{Spec } A_i \rightarrow X$  is an open immersion. As an aside, when  $F \rightarrow G$  is an open immersion, to check whether it is an epimorphism in the big Zariski topos, it suffices to check that  $F(K) \rightarrow G(K)$  is surjective for every field  $K$ .

**3.4. Schemes as locally ringed spaces.** The last approach is the standard construction of schemes as locally ringed spaces. Recall that a locally ringed space is a topological space  $X$  endowed with a commutative ring sheaf  $\mathcal{O}_X$  that is local (in the sense that all the stalks  $\mathcal{O}_{X,x}$  are local rings). Alternatively we have a map of topoi from  $\text{Sh}(X)$  to the Zariski topos (which is the classifying topos of local rings).

The global section functor  $(X, \mathcal{O}_X) \mapsto \Gamma(X, \mathcal{O}_X)$  from locally ringed spaces to  $\text{CRing}^\circ$  has a right adjoint  $A \mapsto \text{Spec } A$  by the general adjoint functor theorem. Since  $\Gamma(\text{Spec } A) = A$ , the functor  $A \mapsto \text{Spec } A$  is fully faithful, so we get an embedding of  $\text{CRing}^\circ$  into the category of locally ringed spaces.

Since the category of locally ringed spaces is cocomplete, we can then use the usual nerve and realisation framework to build a pair of adjoint functors  $|\cdot| \dashv N(\cdot)$ . Here  $N$  is the restricted Yoneda functor  $R \mapsto \text{Hom}(\text{Spec } R, X)$  from locally ringed spaces to presheafs over affine schemes, and  $|cdot|$  is the geometric realization functor built out of the usual coend formula for left Kan extensions: for  $X$  a presheaf seen as a fibration over  $\text{CRing}^\circ$  (via the category of elements construction), its realization is  $|X| = \int^R \text{Spec } R \cdot X_s$  (where  $\cdot$  is the copower).

Then schemes are the locally ringed spaces such that the comonad  $|N(\cdot)|$  is the identity (resp. the presheafs thus that the monad  $N(|\cdot|)$  is the identity); i.e. schemes are identified as the subcategory of locally ringed spaces or presheafs where  $N(\cdot)$ , and  $|\cdot|$  restrict to an equivalence of category [DG70a].

## 4. ELLIPTIC CURVES, ABELIAN VARIETIES AND THEIR POLARISATIONS

### 4.1. Abelian varieties and elliptic curves.

**Definition 4.1.** A field  $k$  (resp. perfect field) is the global sections of a geometrically irreducible and reduced (resp. geometrically reduced) scheme of dimension 0.

A prime  $p$  is non generic point of the small Zariski topos associated to  $\mathbb{Z}$  (i.e. a geometric morphism to  $\text{Set}$ ), and we denote by  $\mathbb{F}_p$  the value of  $\mathcal{O}_{\mathbb{Z}}$  at this point.

**Definition 4.2.** A finite field  $\mathbb{F}_q$  is the global section of a connected finite étale cover of  $\text{Spec } \mathbb{F}_p$ . In particular, we denote by  $\mathbb{F}_{p^2}$  the global section of a connected finite étale cover of  $\text{Spec } \mathbb{F}_p$  of degree 2. All such global sections are isomorphic because the étale fundamental group  $\pi_1^{\text{ét}}(\text{Spec } \mathbb{F}_p, \text{Spec } \overline{\mathbb{F}}_p)$  is procyclic, hence abelian.

A proper morphism  $\pi : X \rightarrow S$  of schemes is a universally closed separated morphism of finite type. A group scheme is a scheme whose functor of points factorize through  $\text{Ab}$ . An abelian scheme  $A/S$  is a proper group scheme.

**Lemma 4.3.** An abelian algebraic space  $A$  over a field  $k$  is an abelian variety.

*Proof.* Trivially obvious from the fact that  $A/k$  is separated, hence a decent algebraic space (i.e. points come from quasi-compact monomorphisms of spectra of fields), which is also a group algebraic space.  $\square$

**Exercise 4.4.** Show that if  $S$  is a scheme, an abelian algebraic space  $A/S$  is an abelian scheme.

If  $\pi : E \rightarrow S$  is an elliptic scheme (i.e. an abelian scheme of relative dimension 1), we let  $\omega_{E/S} = \pi_* \Omega_{E/S}^1$ . A trivialisation of  $\omega_{E/S}$  over an affine open  $\text{Spec } R \subset S$  gives a Weierstrass equation  $E : y^2 + a_1xy + a_3 = x^3 + a_2x^2 + a_4x + a_6$  over  $R$  [KM85]. An elliptic curve  $E/\text{Spec } k$  is an abelian scheme of relative dimension 1 above the spectrum of a field.

**Definition 4.5.** An elliptic curve  $E/\mathbb{F}_{p^2}$  is called supersingular if its crystalline cohomology group  $H_{\text{crys}}^1(E, \mathbb{Z}_{p^2})$  is isomorphic to the  $F$ -crystal  $(\mathbb{Z}_{p^2}^2, \phi)$  where on the basis  $(e_1, e_2)$  we have  $\phi(e_1) = pe_2, \phi(e_2) = e_1$ .



**4.2. Polarisation.** We work in the fppf topos. Recall [Gro72; Bre83] that the stack of biextensions of  $A \times B$  by  $\mathbb{G}_m$  is given by:

$$\tau_{\leq 0} \mathcal{R}Hom(A \otimes^L B, \mathbb{G}_m[1]),$$

that of symmetric biextensions by:

$$\tau_{\leq 0} \mathcal{R}Hom(L \operatorname{Sym}^2 A, \mathbb{G}_m[1]),$$

and that of cubical  $\mathbb{G}_m$ -torsor structures by:

$$\tau_{\leq 0} \mathcal{R}Hom(LP_2 A, \mathbb{G}_m[1]).$$

Here  $P_2$  is the universal quadratic form (for us a quadratic form means of degree  $\leq 2$ ), which admits as a quotient  $\Gamma_2$ , the degree 2 component of the algebra of divided power, which represents the universal homogeneous quadratic form.

By the Dold-Kan correspondance,  $\mathbb{G}_m[1]$  is the delooping  $B\mathbb{G}_m$  of  $\mathbb{G}_m$ ; it can be built as the quotient stack  $[*/\mathbb{G}_m]$ , so  $B\mathbb{G}_m$  is the classifying stack of étale  $\mathbb{G}_m$ -torsors.

In other words, in the internal logic of the derived  $\infty$ -topos, a (symmetric) biextension is a (symmetric) bilinear form  $b$  with values in  $B\mathbb{G}_m$ , and a cubical structure is a quadratic form  $q$  with values in  $B\mathbb{G}_m$ . It is customary to represent a biextension by the symmetric morphism  $\lambda : A \rightarrow \widehat{A}$  associated to it (the biextension on  $A \times A$  is then the pullback by  $1 \times \lambda$  of the canonical Poincaré biextension on  $A \times \widehat{A}$ ). Recall that the dual  $\widehat{A} = \tau_{\leq 0} \mathcal{R}Hom(A, \mathbb{G}_m[1])$  can be seen as the sheaf of linear forms with values in  $B\mathbb{G}_m$ , or equivalently as the morphisms of the Picard stacks from  $A$  to  $B\mathbb{G}_m$ . A  $\Sigma$ -cubical structure is then a “symmetric” quadratic form:  $q(-x) = q(x)$ . (In all that follows, our equalities are internal in the  $\infty$ -logic of the derived fppf topos. In particular,  $a = b$  is an anima rather than a boolean, so our equalities should be witnessed by explicit objects, themselves subject to higher coherence conditions and so on. However, since we work with 1-truncated objects, the coherences condition move from  $B\mathbb{G}_m$  to  $\mathbb{G}_m$  and the recursion stops there.)

A cubical structure  $q : A \rightarrow B\mathbb{G}_m$  gives a symmetric biextension  $b : A \times A \rightarrow B\mathbb{G}_m$  via  $b(x, y) = q(x + y) - q(x) - q(y) + q(0)$ . Forgetting the quadratic structure, the map  $q : A \rightarrow B\mathbb{G}_m$  gives a line bundle  $\mathcal{L}$ ; we define  $K(\mathcal{L})$  the largest subgroup such that  $b$  is trivial over  $K(\mathcal{L}) \times A$ . We let  $\lambda : A \rightarrow \widehat{A}, x \mapsto b(x, \cdot)$  be the associated linear map, then  $K(\mathcal{L}) = \operatorname{Ker} \lambda$ . There are two distinct trivialisations of the symmetric biextension  $b$  over  $K(\mathcal{L}) \times K(\mathcal{L})$ , depending on whether we see  $b$  as the pullback of the canonical Poincaré biextension<sup>4</sup> on  $A \times \widehat{A}$  by  $1 \times \lambda$  or by  $\lambda \times 1$ . They thus differ by a symmetric biextension automorphism, which is precisely the Weil pairing<sup>5</sup>  $e_{\mathcal{L}}$ . For instance, if  $q_0$  is principal, and  $q(x) = q_0(nx)$ , then  $K(\mathcal{L}) = A[n]$ , and the two distinct trivialisations of  $b = nb_0$  amount to whether we use that  $nb_0(x, y) = b_0(nx, y) = 1$  or  $nb_0(x, y) = b_0(x, ny) = 1$  over  $A[n] \times A[n]$ , and the automorphism between the two trivialisations is given by the Weil pairing  $e_n : A[n] \times A[n] \rightarrow \mu_n$ .

Also, the Tate pairing is induced by the trivialisation of  $b$  above  $K(\mathcal{L}) \times A$ . For instance, when  $q(x) = q_0(nx)$  for  $q_0$  principal, then if  $P \in K(\mathcal{L})(k) = A[n](k), Q \in A(k)$ , the Tate pairing  $e_{T, \mathcal{L}}(P, Q)$  is the étale  $\mu_n$ -torsor measuring the obstruction of lifting  $P$  to the slice  $\mathcal{L} \mid A[n] \times \{Q\}$  in a way compatible with its associated squared structure.

Furthermore, over  $K(\mathcal{L}) \times K(\mathcal{L})$ ,  $b$  is trivial, so  $q$  is linear:  $\mathcal{L} \mid K(\mathcal{L})$  has a squared structure, and  $q : K(\mathcal{L}) \rightarrow B\mathbb{G}_m$  is a morphism of Picard stacks, so gives an element of  $\operatorname{Ext}^1(K(\mathcal{L}), \mathbb{G}_m)$ , i.e. a group extension  $G(\mathcal{L})$  of  $K(\mathcal{L})$  by  $\mathbb{G}_m$ .  $G(\mathcal{L})$  is called the theta group of  $\mathcal{L}$ . Lastly, since  $b$  is trivial over  $K(\mathcal{L}) \times A$ ,  $q(x + y) + q(0) = q(x) + q(y)$  for all  $x \in K(\mathcal{L})$  and  $y \in A$ , which give an action of  $\mathcal{L} \mid K(\mathcal{L})$  on  $\mathcal{L} \mid A$ ; this is the theta group action.

**Example 4.6.** Let  $q : A \rightarrow B\mathbb{G}_m$  be a quadratic form and  $\mathcal{L}$  the line bundle. Then  $q$  is linear iff  $K(\mathcal{L}) = A$  iff  $\mathcal{L}$  has a squared structure over  $A$ , iff  $G(\mathcal{L}) \in \operatorname{Ext}^1(A, \mathbb{G}_m)$ , iff  $\mathcal{L} \in \operatorname{Pic}^0(A)$ , and we recover the isomorphism  $\widehat{A} \simeq \operatorname{Pic}^0(A)$ .

**Definition 4.7** ((Symmetric) Theta structure). Given a positive cubical (resp.  $\Sigma$ -cubical) structure  $q : A \rightarrow B\mathbb{G}_m$ , with associated (symmetric) line bundle  $\mathcal{L}$  then a (symmetric) theta structure is:

- A symplectic isomorphism of  $(K(\mathcal{L}), e_{\mathcal{L}})$  with  $\prod_{i=1}^g \mathbb{Z}/d_i \mathbb{Z} \times \prod_{i=1}^g \mu_{d_i}$  with its canonical pairing (recall that  $\mu_d$  is the Cartier dual of  $\mathbb{Z}/d\mathbb{Z}$ ).

<sup>4</sup>Associated to the quadratic form  $q : (\phi, x) \mapsto \phi(x)$

<sup>5</sup>The Weil pairing is alternate, because obviously the automorphisms of a trivial symmetric biextension over  $X \times X$  are given by alternate bilinear forms  $X \times X \rightarrow \mathbb{G}_m$ . Conversely, the automorphisms of the trivial antisymmetric (resp. alternate) biextension over  $X \times X$  are given by symmetric bilinear forms  $X \times X \rightarrow \mathbb{G}_m$  (resp. symmetric bilinear forms induced by a quadratic form  $X \rightarrow \mathbb{G}_m$ ) [Bre87].

- A choice of trivialisation of the cubical (resp.  $\Sigma$ -cubical) structure above  $K_1(\mathcal{L})$  and  $K_2(\mathcal{L})$ .

**Corollary 4.8.** *A symmetric theta structure is defined over  $k$  if the isomorphism  $K(\mathcal{L}) \simeq \prod_{i=1}^8 \mathbb{Z}/d_i\mathbb{Z} \times \prod_{i=1}^8 \mu_{d_i}$  is rational, and if the 2g étale  $\mu_{2 \wedge d_i}$ -torsors that control the trivialisation of the  $\Sigma$ -cubical structure are trivial over  $k$ . (These étale torsors can be described as a slight generalisation of the Tate pairing, using the monodromy interpretation of the Tate pairing in biextensions).*

Given a cubical structure  $q : A \rightarrow B\mathbb{G}_m$  with associated line bundle  $\mathcal{L}$  and symmetric biextension  $b$ , and  $\phi : A \rightarrow B$  a morphism with kernel  $K$ , then  $q$  descends to  $B$  if and only if  $b(x+z, y) = b(x, y)$  for all  $z \in K, x, y \in A$ , and  $q(z) = q(0)$  for all  $z \in K$ , i.e. we have a trivialisation of the biextension over  $K \times A$  and of the cube structure over  $K$  (in a compatible way).

In particular, given a theta structure, we obtain by definition a descent of  $q$  to  $A/K_2(\mathcal{L})$ , which gives by pullback a section  $\theta_0$ . Now we use the theta group action along with the trivialisation above  $K_1(\mathcal{L})$  to obtain an action of  $K_1(\mathcal{L})$  on  $\theta_0$  which defines the remaining theta functions  $\theta_i$ .

As an other application, if  $\phi : A \rightarrow B$  is an isogeny, and we picked a choice of trivialisation of  $q$  above  $K = \text{Ker } \phi$ , then the sections on  $B$  (for the descent of  $q$ ) are exactly the section of  $A$  invariant under the action of  $K$  induced by this trivialisation.

**Corollary 4.9.** *One can compute an  $\ell$ -isogeny  $(A, \lambda_A) \rightarrow (B, \lambda_B)$  in polynomial time  $O(\ell^8)$ .*

## 5. MAXIMAL SUPERSINGULAR ELLIPTIC CURVES AND QUATERNION ALGEBRAS

**5.1. Quaternion algebras.** Recall that the  $K$ -theory functor  $K$  is the universal additive invariant from compactly generated stable  $\infty$ -categories to  $\text{Sp}$  (the category of spectra); in particular for any other such invariant  $E$  we have  $E(\mathbb{S}) = \text{Nat}(K, E)$ , where  $\mathbb{S}$  is the sphere spectrum.

For  $R$  a ring spectrum, we let  $K(R) = K(\text{Perf}(R))$ , where  $\text{Perf}(R)$  is the category of perfect (i.e. compact) animated  $R$ -modules. In particular for a ring  $R$ ,  $K(R) \simeq K(HR)$  is the spectrum  $K(\text{Perf}(R))$  where  $\text{Perf}(R)$  is the  $\infty$ -category of derived perfect complexes over  $R$  and  $HR$  is the Eilenberg-MacLane spectrum associated to  $R$ .

Now for  $R = F$  a field of characteristic 0 or prime to  $m$ , by the motivic Bloch–Kato conjecture, we have a norm residue isomorphism  $K_2(F)/mK_2(F) \simeq H_{\text{et}}^2(F, \mu_m^{\otimes 2})$ , where  $K_2 = \pi_2 \circ K$  and  $\pi_2 = \pi_0 \circ \Omega^2$  is the second homotopy group. In particular, if  $\mu_m \subset K$ , all classes in  $Br(F)[m] \simeq H_{\text{et}}^2(F, \mu_m)$  come from a cyclic Azumaya algebra.

**Definition 5.1.** A quaternion algebra over  $\mathbb{Q}$  is an Azumaya algebra whose class in the Brauer group lands in (the image of)  $H_{\text{et}}^2(\mathbb{Q}, \mu_2)$ .

Equivalently, given a Severi–Brauer variety  $X$  of dimension 2 over  $\mathbb{Q}$ , with associated étale  $\text{PGL}_2(\mathbb{Q})$ -torsor  $Y = \text{Isom}(\mathbb{P}_{\mathbb{Q}}^2, X)$ , we obtain a quaternion algebra by applying this torsor to  $M_2(\mathbb{Q})$ .

**5.2. Weil’s restriction.** Let  $E_0/\mathbb{F}_{p^2}$  be a maximal supersingular elliptic curve with endomorphism ring  $O_0$ . There is an obvious left  $O_0$ -module contravariant action on  $E_0$ . Let  $\pi : \text{Spec } \mathbb{F}_{p^2} \rightarrow \text{Spec } \mathbb{F}_p$  be the natural map, then the Weil restriction on étale abelian sheaves is the pushforward map  $\pi_*$  from the topos of étale sheafs over  $\text{Spec } \mathbb{F}_{p^2}$  to the topos of étale sheafs over  $\text{Spec } \mathbb{F}_p$ . Since  $\pi$  induces a geometric morphism  $(\pi^{-1}, \pi_*)$  of topos,  $\pi_*$  is right adjoint to  $\pi^{-1}$ ; since in our case  $\pi$  is finite étale,  $\pi_* = \pi_!$  is also a left adjoint of  $\pi^{-1}$ . We call the functor  $\pi_*$  the Weil restriction  $W$ .

In particular we have a comonad  $\text{Tr} = \pi_*\pi^{-1}$ , called the trace comonad. For instance, if  $E/\mathbb{F}_p$  is rational, then the counit gives the usual trace map  $WE(\mathbb{F}_p) = E(\mathbb{F}_{p^2}) \rightarrow E(\mathbb{F}_p), P \mapsto P + P^\sigma$ .

**Theorem 5.2.** *Assume that  $E_0/\mathbb{F}_p$  is a supersingular curve, then its base change to  $\mathbb{F}_{p^2}$  is maximal supersingular. Let  $R = \text{End}_{\mathbb{F}_p}(E_0)$  and  $O_0 = \text{End}_{\mathbb{F}_{p^2}}(E_0)$ , Let  $M$  be a left  $O_0$ -module, then  $W(M \cdot_{O_0} E_0) = M \cdot_R E_0$ . More generally, for  $M$  a left animated  $O_0$ -module,  $R\pi_* R\mathcal{H}om_{O_0}(M, E_0) = R\mathcal{H}om_R(M, E_0)$ .*

*Proof.* Obviously trivial from the isomorphism  $Rf_* R\mathcal{H}om_{D(O_X)}(Lf^*(F), G) \simeq R\mathcal{H}om_{D(O_Y)}(F, Rf_*(G))$  for any geometric morphism  $(f^*, f_*) : (X, O_X) \rightarrow (Y, O_Y)$  of ringed topoi.  $\square$

Thanks to [Theorem 5.2](#), we can move from quaternionic module actions to quadratic module actions via the Weil restriction.

**5.3. The module action on polarisations.** A subtlety compared to the general purpose NIKE in Definition 2.2 is that we need to keep track of polarisations.

Fix  $E_0/\mathbb{F}_p$  a supersingular elliptic curve, and let  $R = \text{End}_{\mathbb{F}_p}(E_0)$ . Let  $A/\mathbb{F}_p$  be isogeneous to  $E_0^g$ .

Thanks to Section 4.2 this is easy. Take  $b : A \times A \rightarrow B\mathbb{G}_m$  a symmetric biextension, and let  $\lambda : A \rightarrow \widehat{A}$  be the associated polarisation. Take  $M$  an  $R$ -module and  $H$  a unimodular  $R$ -Hermitian form  $H : M \times M \rightarrow R$ , and let  $\lambda_H : M^\vee \rightarrow M$  the associated polarisation.

Define a biextension on  $M \cdot A = \mathcal{R}\text{Hom}_R(M, A)$  via  $\phi \in \mathcal{R}\text{Hom}_R(M, A) \mapsto \lambda \circ \phi \circ \lambda_H$  (all this in the internal logic of the  $\infty$ -topos). This defines a polarisation  $M \cdot A \rightarrow M^\vee \cdot A^\vee = (M \cdot A)^\vee$ , where the last equality follows from

$$\mathcal{R}\text{Hom}_R(\mathcal{R}\text{Hom}_R(M, R), \mathcal{R}\text{Hom}(A, B\mathbb{G}_m)) = \mathcal{R}\text{Hom}(\mathcal{R}\text{Hom}_R(M, A), B\mathbb{G}_m).$$

## 6. MIKE

Now the description of MIKE should be obvious from Sections 2 and 5, and how to implement it follows easily from Section 4.

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#### APPENDIX A. IMPLEMENTATION

We provide a Cobol implementation of MIKE. (We acknowledge the help of LLM to convert the Sage implementation made by the MIKE team to Cobol). We let to the reader the trivial exercise of completing the implementation, notably the 4d isogeny part.

##### IDENTIFICATION DIVISION.

```
PROGRAM-ID. MIKE-POST-QUANTUM-KEY-EXCHANGE.
AUTHOR. SAGE-DEFECTOR.
DATE-WRITTEN. 01-APR-2026.
SECURITY. UNCLASSIFIED-BUT-DEEPLY-CONFUSING.
```

##### ENVIRONMENT DIVISION.

##### CONFIGURATION SECTION.

```
SOURCE-COMPUTER. IBM-MAINFRAME-FROM-1974.
OBJECT-COMPUTER. IBM-MAINFRAME-FROM-1974.
```

##### DATA DIVISION.

##### WORKING-STORAGE SECTION.

```
* =====
* CRYPTOGRAPHIC PARAMETERS (HOPING THEY FIT IN 99 DIGITS)
* =====

01 MIKE-PARAMETERS.
   05 WS-PARAM-NAME          PIC X(10) VALUE "TOY".
   05 WS-PRIME-P             PIC 9(99).
   05 WS-VALUATION-F         PIC 9(4).
   05 WS-EXPONENT-E          PIC 9(4).
   05 WS-ODD-COFACTOR        PIC 9(99).

01 ELLIPTIC-CURVE-DATA.
   05 WS-X-COORD             PIC 9(99)V9(10).
   05 WS-Y-COORD             PIC 9(99)V9(10).
```

```

05 WS-MONTGOMERY-A          PIC S9(99).
05 WS-MONTGOMERY-B          PIC S9(99) VALUE 1.

01 SECRET-KEY-DATA.
05 WS-SECRET-SCALAR-X      PIC 9(99).
05 WS-ALPHA-N              PIC S9(99).
05 WS-SECRET-A             PIC S9(99).
05 WS-SECRET-B             PIC S9(99).

01 THETA-STRUCTURE-NIGHTMARE.
05 WS-IS-SYMPLECTIC        PIC X(1) VALUE 'N'.
    88 MATRIX-IS-SYMPLECTIC VALUE 'Y'.
05 WS-SCHOLTEN-ISOGENY     PIC X(50) VALUE "MATH-TOO-HARD-FOR-COBOL".

01 ENTROPY-SOURCE.
05 WS-DRBG-SEED            PIC X(48).
05 WS-RANDOM-BYTES         PIC X(256).

```

# PROCEDURE DIVISION.

## 000-MAIN-LOGIC.

```

    DISPLAY "INITIALIZING MIKE POST-QUANTUM KEY EXCHANGE...".
    DISPLAY "PLEASE INSERT PUNCH CARDS FOR FINITE FIELD ARITHMETIC.".

    PERFORM 100-SET-PARAMETERS-FROM-NAME.
    DISPLAY "MAPPING Q ABOVE (0,0) TO ENSURE P+[x]Q NEVER HITS 0.".
    PERFORM 200-KEYGEN-ALICE.
    PERFORM 300-DERIVE-SHARED-SECRET.

    DISPLAY "KEY EXCHANGE COMPLETE. HOPE YOU DID NOT DIVIDE BY ZERO.".
    STOP RUN.

```

## 100-SET-PARAMETERS-FROM-NAME.

```

    EVALUATE WS-PARAM-NAME
        WHEN "TINY"
            MOVE 8 TO WS-VALUATION-F
            COMPUTE WS-PRIME-P = (2 ** 8 * 5) - 1
        WHEN "TOY"
            MOVE 32 TO WS-VALUATION-F
            COMPUTE WS-PRIME-P = (2 ** 32 * 5) - 1
        WHEN "NIST_ONE"
            MOVE 248 TO WS-VALUATION-F
            COMPUTE WS-PRIME-P = (2 ** 248 * 5) - 1
        WHEN OTHER
            DISPLAY "ERROR: PARAMETER NOT SUPPORTED."
            STOP RUN
    END-EVALUATE.
    COMPUTE WS-EXPONENT-E = WS-VALUATION-F - 3.

```

## 200-KEYGEN-ALICE.

```

    DISPLAY "SAMPLING SECRET SCALAR...".
    * Pretend we have os.urandom in 1960
    MOVE "RANDOM-GARBAGE" TO WS-RANDOM-BYTES.
    MOVE 42 TO WS-SECRET-SCALAR-X.

    DISPLAY "COMPUTING PRIVATE KEY [ALPHA, A, B]...".
    COMPUTE WS-ALPHA-N = (WS-SECRET-SCALAR-X * 3 - 1) ** 2.

    DISPLAY "COMPUTING PUBLIC KEY VIA ISOGENY...".

```

PERFORM 250-COMPUTE-MONTGOMERY-MODEL.

250-COMPUTE-MONTGOMERY-MODEL.

\* s = 1 / (3\*alpha^2 - 1) -> Good luck doing modular inverse here  
COMPUTE WS-MONTGOMERY-A = -3 \* 1.

300-DERIVE-SHARED-SECRET.

DISPLAY "EVALUATING SCHOLTEN CONSTRUCTION...".

DISPLAY "APPLYING WEIL RESTRICTION. CONJUGATING CURVE...".

IF WS-IS-SYMPLECTIC = 'N'

DISPLAY "CRITICAL ERROR: BASIS IS NOT SYMPLECTIC."

DISPLAY "PLEASE CALL ALGEBRAIC GEOMETER FOR SUPPORT."

ELSE

DISPLAY "COMPUTING GLUING CHANGE OF COORDINATES..."

DISPLAY "COMPUTING 4D ISOGENY..."

DISPLAY "WARNING: COMPUTING DIMENSION 4 ISOGENY IN COBOL MAY".

DISPLAY "CAUSE MAINFRAME TO CATCH FIRE..."

DISPLAY "SHARED SECRET DERIVED SUCESSFULLY."

END-IF.

1 RUE DU POISSON D'AVRIL

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